

Acids and Bases

Arrhenius and Bronsted-Lowry Definitions

According to the Arrhenius definition, an acid increases the concentration of hydrogen ions in aqueous solution, while a base increases the concentration of hydroxide ions.

The Bronsted-Lowry definition is broader: an acid is a proton donor and a base is a proton acceptor, which allows the definition to apply beyond aqueous solutions.

pH Scale

The pH scale measures the acidity of a solution and is calculated as the negative base-10 logarithm of the hydrogen ion concentration: $\text{pH} = -\log[\text{H}^+]$.

A pH of 7 is neutral, values below 7 are acidic, and values above 7 are basic, with each whole number step representing a tenfold change in hydrogen ion concentration.

Strong vs Weak Acids

A strong acid fully dissociates into ions in water, while a weak acid only partially dissociates, establishing an equilibrium between the acid and its ions.